Understanding Methane and Carbon Dioxide Fluxes in the Mackenzie River Delta

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The Mackenzie River Delta in the Northwest Territories is the Arctic's second largest river delta and is of particular concern with regard to methane emissions because methane "hotspots" have been detected by multiple aerial surveys, which are attributable to geologic methane seeps and/or to biologic methane emissions from its expansive wetland areas, both of which have the potential to increase with permafrost degradation due to climate change[1,2,3,4].

Methodology

Eddy covariance (EC) is one of the primary methods to monitor landscape-scale greenhouse gas exchange and understand the dynamics controlling gas exchange over time. To reduce the uncertainties regarding the trajectory of greenhouse gas fluxes in the delta, an EC system was installed in mid-summer 2024 at a sedge wetland site underlain by thin continuous permafrost which is representative of the outer delta area. The system measures the ecosystem-atmosphere exchange of methane and carbon dioxide and collects ancillary data that include solar radiation, air temperature, soil temperature profiles, and water table depth. These data are used to understand the functional relationships driving greenhouse gas fluxes in the region and to estimate the greenhouse gas balance of the region. The 2024 field season was an exploratory campaign, limited from the mid-July to mid-September using borrowed EC instrumentation. A longer-term, multi-year campaign is planned to begin in June 2025 with the installation of newly procured EC sensors.

Results

This work is in its early stages and results to date are limited (Fig 1). The site is intended to provide a multi-year time series of methane and carbon dioxide fluxes for an ecosystem representative of the region. The primary long-term objectives of the station are: 1) to understand non-growing season fluxes, with particular focus on the snowmelt and freeze-up periods, 2) to estimate the ecosystem's net greenhouse gas balance by accounting for the disproportionally high global warming potential of methane fluxes, and 3) to serve as an anchor point from which further studies on the spatial variability and magnitude of methane emissions in the outer Mackenzie River Delta can be based. An overview of preliminary findings from the 2024 deployment will be presented along with the study design for upcoming seasons that include the use of flux chambers to define contributions from various distinct landscape units, and a discussion of this work against the broader context of permafrost degradation and implications on feedbacks to climate change.

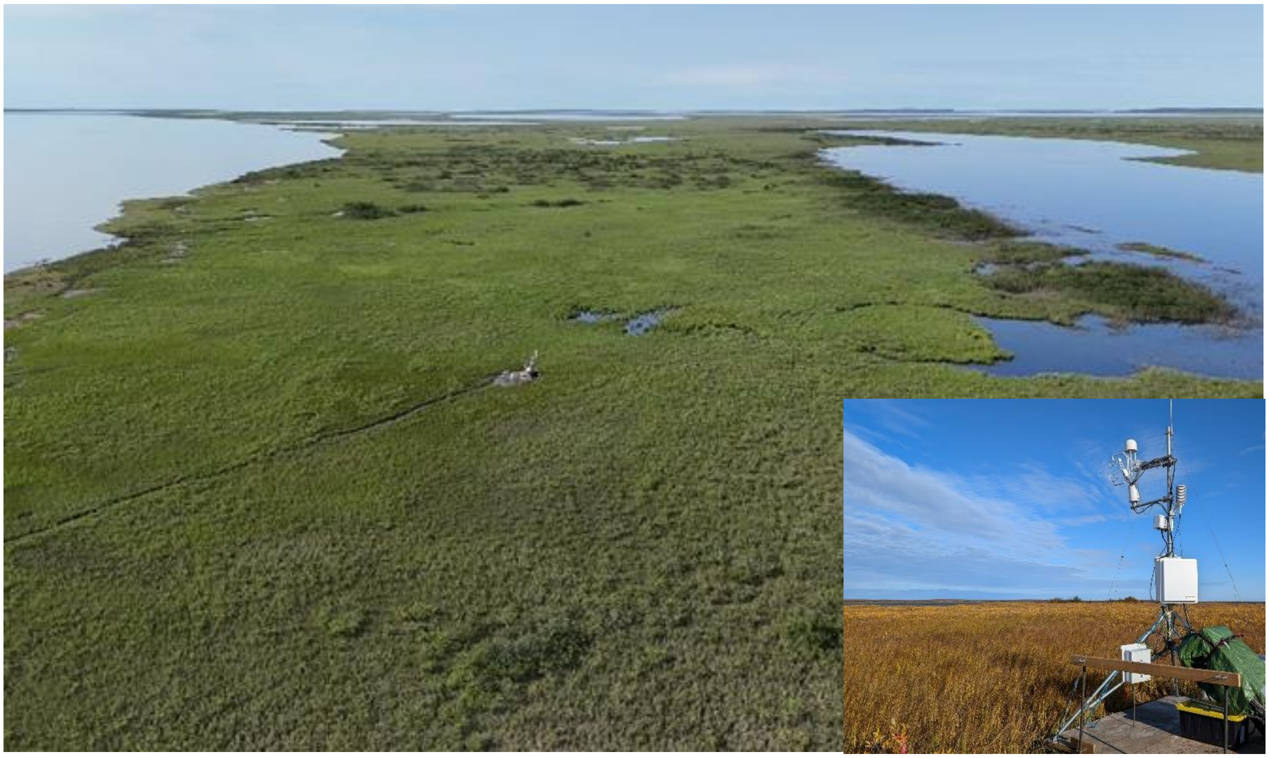

Fig1. The main photo shows oblique drone imagery of the Eddy Covariance station within the broader landscape at the beginning of the 2024 campaign (July 19) and the inset in the bottom right shows the station at the end of the 2024 campaign mid-Autumn (September 12th). The station is located at N 69.226, W 135.252 approximately 10 km southeast of the Arctic Ocean.

References

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